

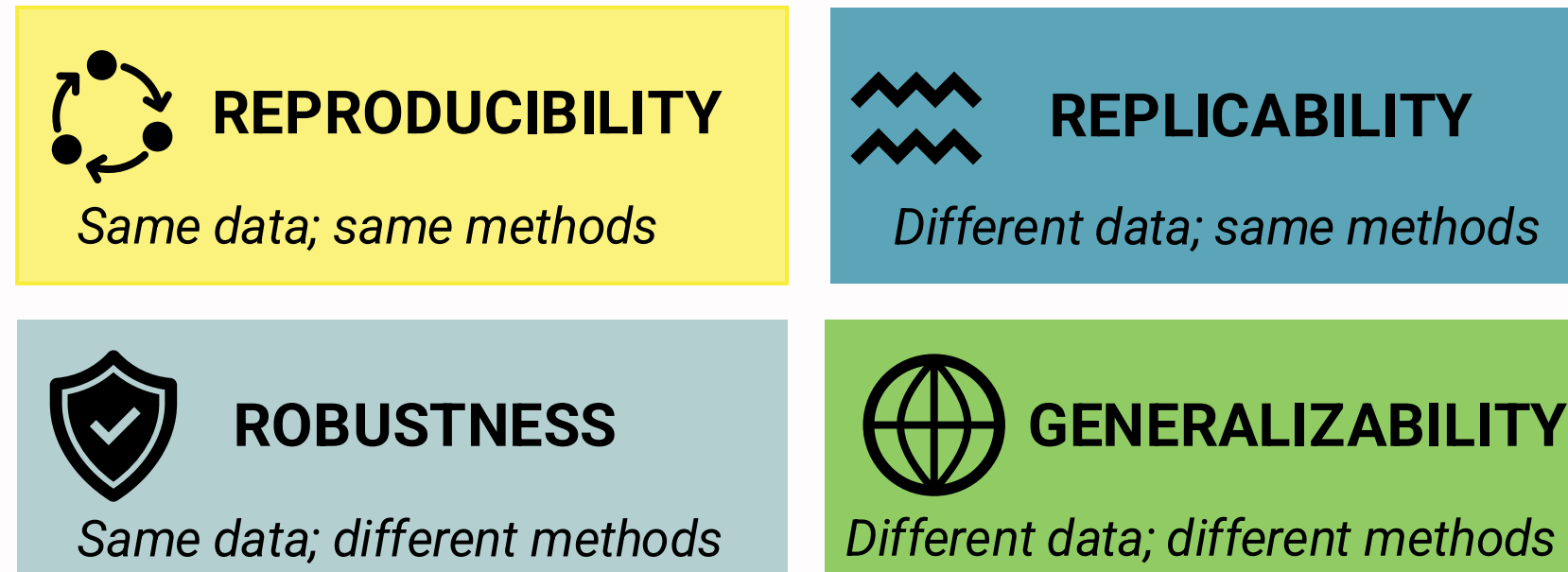
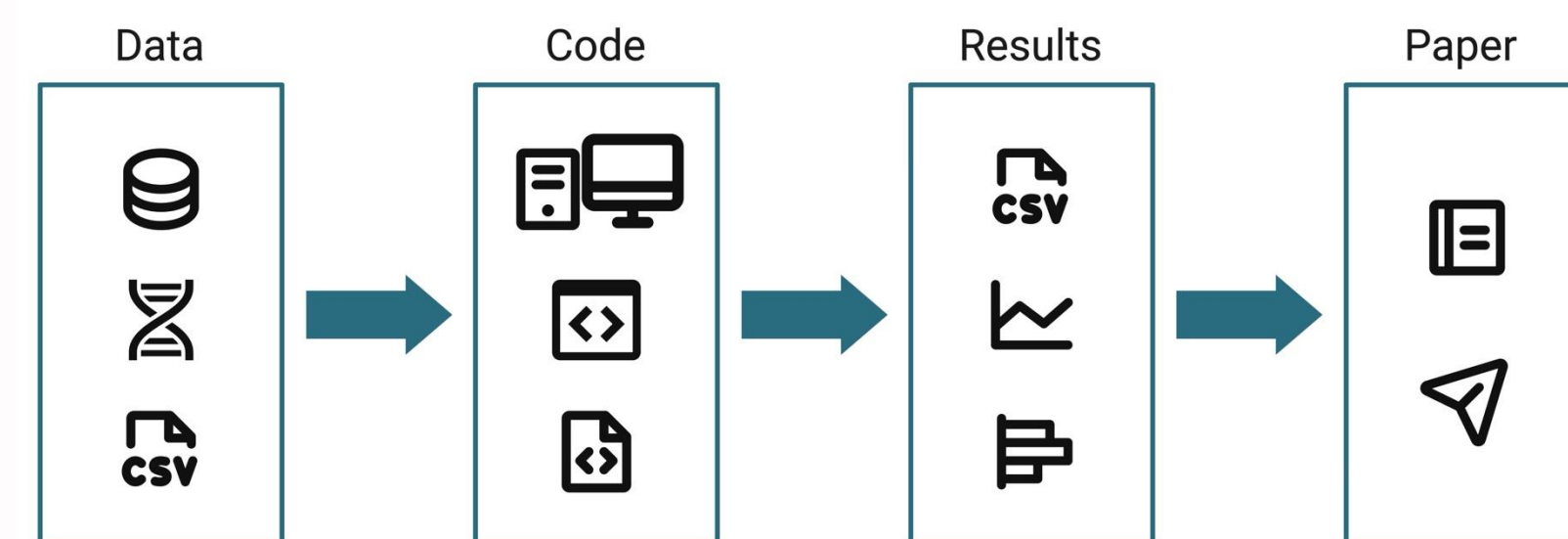
# Anatomy Of A Modular Toolchain For Reproducible Bioinformatics Workflows

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## INTRODUCTION

Reproducibility is the foundation of trustworthy computational biology. Our framework operationalizes core principles across all pipelines



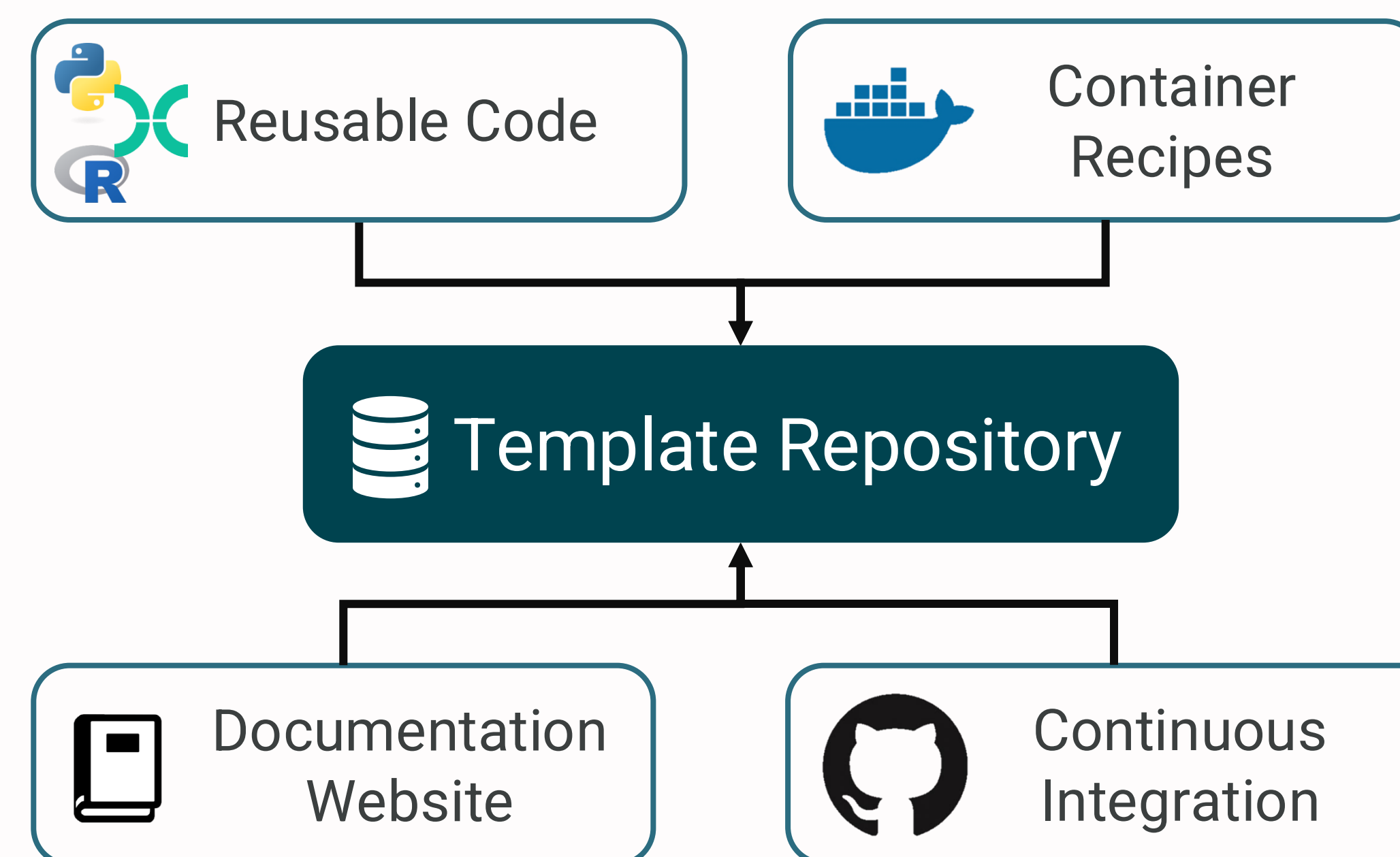
Schloss PD. 2018. 10.1128/mBio.00525-18.

## BEST PRACTICES

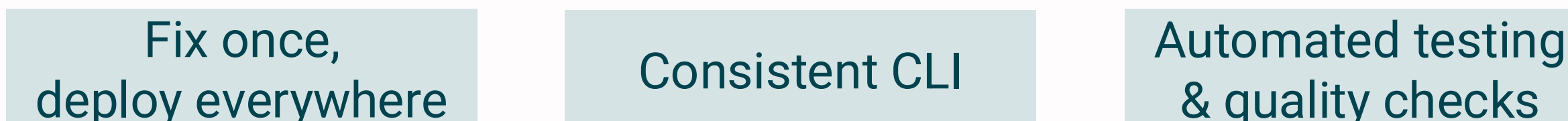


We enforce consistency through templates and automation. These practices reduce technical debt and improve maintainability.

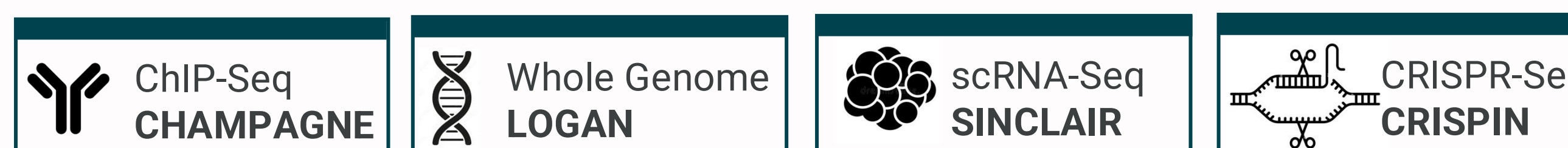
## TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



Unified framework from template to deployment with automated quality and delivery

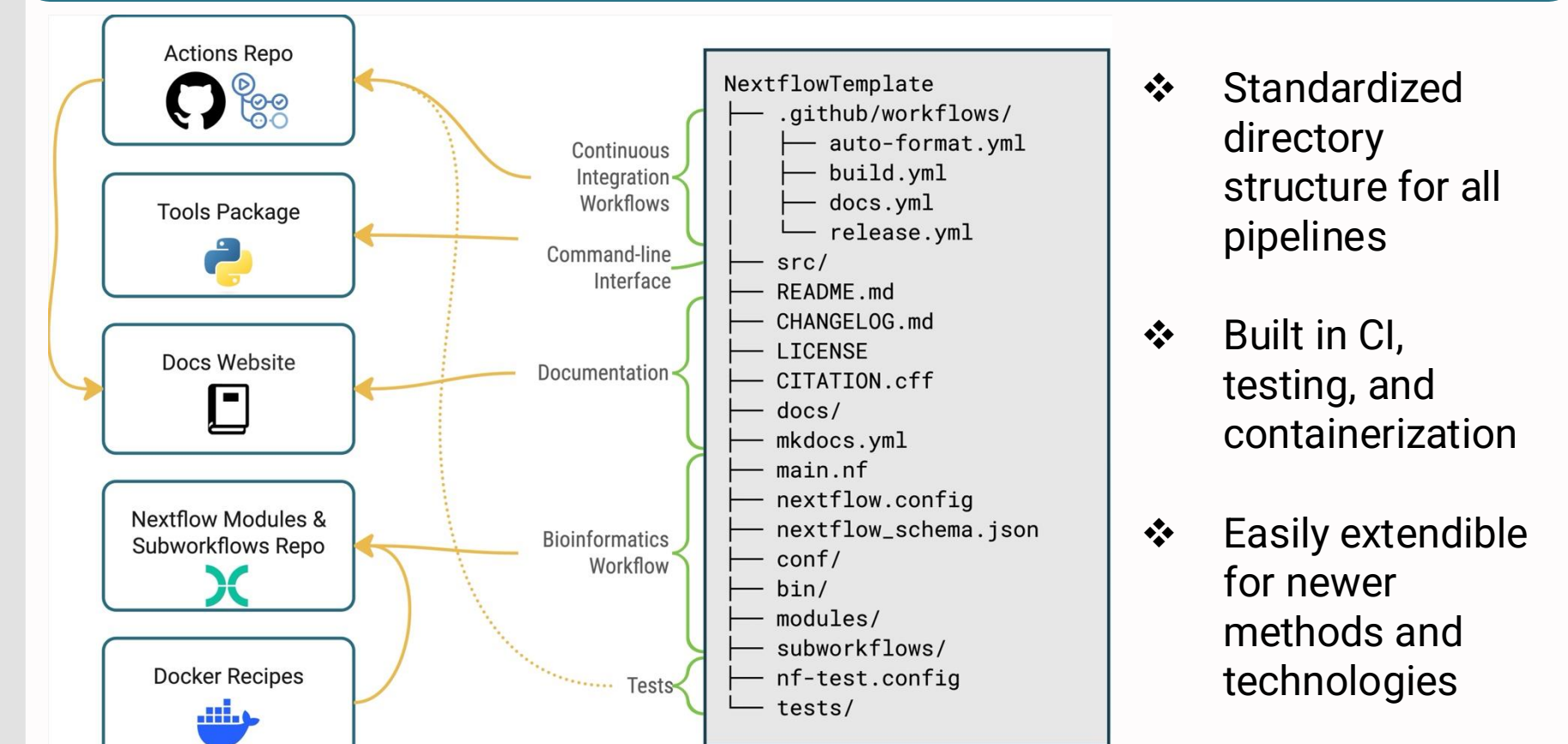


## PIPELINES BUILT



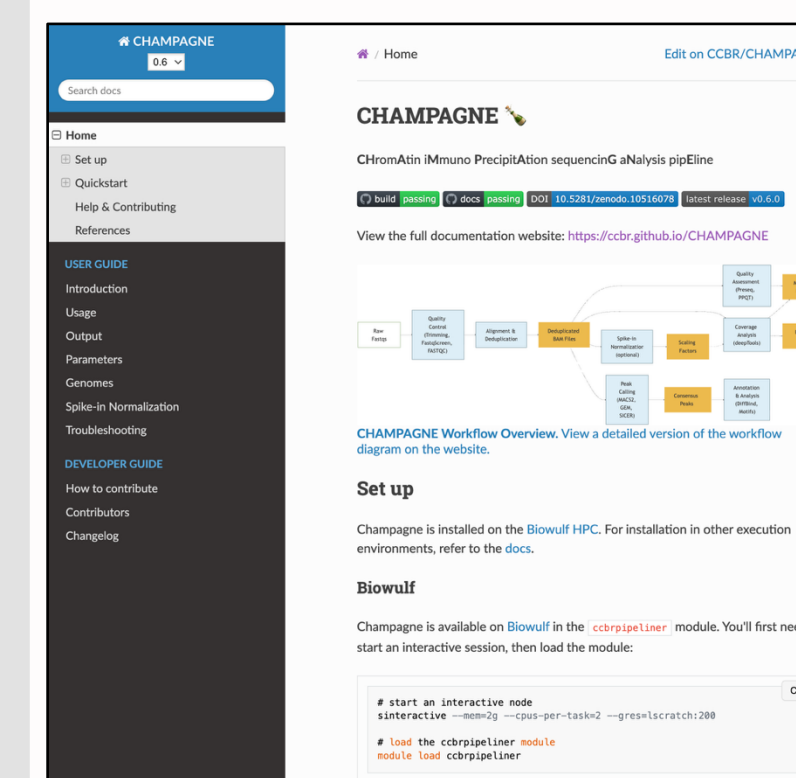
**10+** pipelines on NIH Biowulf HPC cluster  
\$ module load ccbripipeline

## PIPELINE TEMPLATE



- ❖ Standardized directory structure for all pipelines
- ❖ Built in CI, testing, and containerization
- ❖ Easily extendible for newer methods and technologies

## CLI, DOCUMENTATION, AUTOMATION



\$ champagne run --input samplesheet  
\$ Sinclair run --input samplesheet

Consistent usage and documentation across all pipelines simplifies user training

GitHub Actions runs CI checks on every change

## Scalable, Reproducible, Maintainable Bioinformatics

